



Hacettepe Hospital *Indoor* Navigation.

An LLM-assisted, graph-based wayfinding system for complex hospital environments.

TEAM

Mert Çetin · Azra Sena Kansu
Bora Dere · Berkay Yıldız

SUPERVISOR

Prof. Dr. Mehmet Önder Efe

COURSE

BBM480 — Final Defense
Group 53

EXHIBITION

May 15, 2026 · 14:00
Computer Engineering Bldg

Lost in a *complex* building.

8

BLOCKS

5

FLOORS

352

NAMED LOCATIONS

01

THE PROBLEM

Hacettepe University
Hospital spans 8 blocks and
5 floors. Indoors, GPS fails.
Signage is static. The
human cost is real.

SCALE

352 named locations across 8
blocks. Patients arrive without
knowing which entrance to take.

STATIC SIGNAGE

Fixed signs do not adapt to
closures, queues, or accessibility
constraints.

HUMAN COST

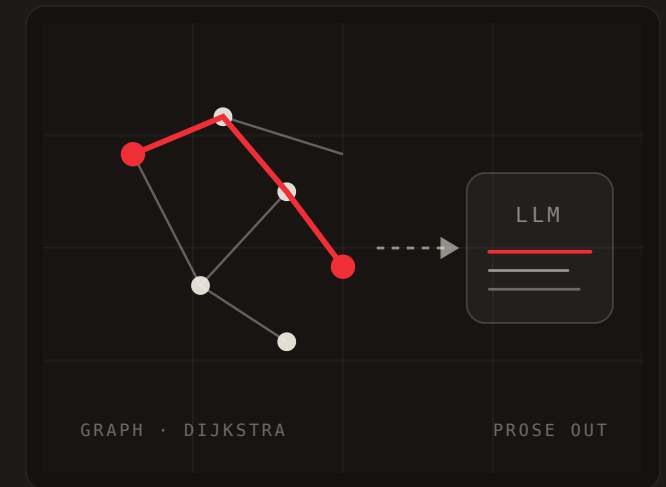
Reception desks absorb the gap.
Wheelchair users walk extra
floors by mistake.

Can *deterministic graph-search + schema-constrained LLMs* produce navigation that is *provably correct, fluent, landmark-anchored, and accessibility-aware* — *without specialized hardware*?

02

RESEARCH QUESTION WHY IT'S HARD

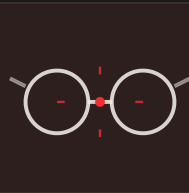
- 01 Floor plans rarely encode walkable connectivity directly.
- 02 Free-form LLM output can hallucinate hallways that don't exist.
- 03 Accessibility constraints must propagate through routing, not just UI.
- 04 Instructions must read like a human would describe the route.



CONCEPT DIAGRAM

Four research families.

Indoor navigation sits at the intersection of four streams. We borrow from each and unify them around natural-language output.



AR + VIO

AR overlays + visual-inertial odometry for inside-out localization. Strong UX, expensive sensors.

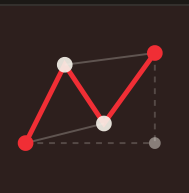
Helan et al.
· Rehman et al.



ACCESSIBILITY-AWARE ROUTING

Weighting elevators, ramps, and stair-free paths for wheelchair and limited-mobility users.

Kumar et al. · Ng et al.



ALGORITHMIC CORE

Dijkstra, A*, contraction hierarchies on weighted indoor graphs — deterministic and auditable.

Saha & Banerjee · Google Maps



LLMS FOR SPATIAL REASONING

Rewriting raw turn lists into natural prose; constrained decoding to keep facts grounded.

Jiang et al. · Yamada et al. · Mansourian et al.

OUR
GAP

Integrate all four into one system grounded in a real hospital.

03

STATE
OF THE
ART

Three layers, one REST boundary.

A vertical pipeline. Each layer owns one job. The data layer never talks to the LLM directly.

L1

Presentation Layer

Next.js client + FastAPI gateway.
Strict JSON contract over /api/route.

frontend

L2

Routing + NLU Engine

Dijkstra over the weighted graph, then MiniMax-M2.7 rewrites the path into natural Turkish.

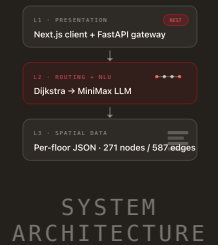
backend

L3

Spatial Data Layer

Per-floor JSON graphs: 271 nodes, 587 edges, manually placed against architectural plans

data

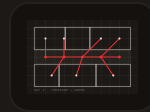


From floor plans to graph.

05 SPATIAL
DATA

PROCESS

- 01 **INPUT** .dwg architectural drawings.
- 02 **EXPORT** PDF per floor via Autodesk DWG TrueView.
- 03 **ANNOTATE** Manual node + edge placement (e.g. KAT_1_19_1 asansör node).
- 04 **EMIT** Per-floor JSON, schema-validated.



ANNOTATED
FLOOR
PLAN →
GRAPH

271

NODES

587

EDGES

4

FLOORS

271 NAVIGABLE
LOCATIONS ACROSS
4 FLOORS · EDGE
MIX: CORRIDOR
88.7% · STAIRS
7.5% · ELEVATOR
3.8%

Dijkstra finds it. The LLM *tells* it.

06

ROUTING

PATHFINDING

Dijkstra on a small, weighted graph.

```
weight(edge) = base_cost
if avoid_stairs and edge.kind == 'stairs':
    weight += 10^9    # near-infinite soft penalty
```

- Graph is small (≤ 300 nodes); A* heuristic gains nothing.
- Deterministic — same input, same path. Easy to audit.
- Accessibility weights compose without re-architecting.

INSTRUCTION
REWRITING

MiniMax-M2.7
with strict JSON
schema.

Python

FastAPI

Dijkstra

MiniMax LLM

JSON Schema

IN raw turn list
— e.g.
KAT_1_10
→ 19_1
(asansör)
→
KAT_2_07

SCHEMA JSON
schema with
enum
landmarks;
no free fields

CLEAN strip
<think>...
</think>
traces
before

From engine to thumb.

A REST API with three honest endpoints, and a Next.js client that looks like it was designed by a hospital — not a startup.

07

INTERFACE

API SURFACE

GET /health →
liveness probe

GET /api/nodes
→ flat list of
named locations

POST /api/route
→ {start, goal,
avoid_stairs} →
route + prose

SCREEN FLOW

Next.js

01 **Start** type-ahead with
Turkish diacritics

TypeScript

Framer
Motion

02 **Destination** merdivensiz
rota toggle

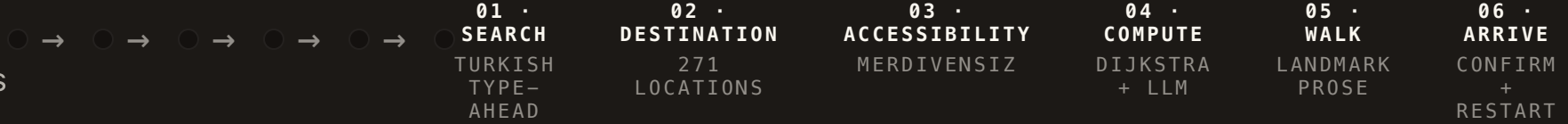
Expo
(legacy)

03 **Navigation** step-by-
step with landmarks

04 **Arrival** confirm + restart

Six steps, one journey.

Real
screenshots
from the
deployed
Next.js
client.
Search →
toggle →
compute →
walk →
arrive.



08

HOW IT
WORKS



FIELD
STUDY

HACETTEPE
HOSPITAL ·
MAY 2026 ·
N ≈ 10

What people told us.

Informal pilot on-site with patients and hospital team members. Small sample, sharp signal.

LIKED

- Step-by-step logic was easy to follow.
- UI felt intuitive without instructions.
- Turkish prose felt natural, not translated.
- Reached destinations on the first try.

RAISED

- Unclear where I'm standing — start point selection felt risky.
- What if I pick the wrong starting node?
- Floor change instructions could be more visual.

*Validates the need for **auto-localization**.*

QR codes

NFC

BLE beacons

What the *numbers* say.

10

RESULTS

8 pytest
scenarios,
all green.
100
randomly
sampled
routes.
LLM
parsing
under
contract.

8

PYTEST
SCENARIOS

98%

ROUTING
SUCCESS

0.36ms

MEDIAN LATENCY

14.3

MEAN STEPS

≈ 51S EST.
WALKING TIME
· N = 100
ROUTES ·
ACCESSIBILITY
MODE SHIFTS
35 ROUTES
STAIRS →
ELEVATOR · N
INVENTED
DEPARTMENTS

Beyond *one* hospital.

The architecture generalizes
to any indoor environment with
a graph and a story to tell.

IMMEDIATE IMPACT

- Self-service wayfinding reduces reception load.
- Accessibility toggle is one switch, not a separate flow.
- Turkish-first instructions match how patients actually ask.

NEXT STEPS

- Annotate Floor 4 to close the building.
- STT/NLU front-end: speak your destination.
- AR overlay + visual positioning (anchors).
- Real-time disruption signals (closures, queues).
- Formal user study against signage baseline.

Questions?



hacettepehospital.vercel.app

11 WHAT'S
NEXT

MAY 15 · 2026 · 14:00 ·
CMP ENG BLDG